FIELD OPERATIONS

# Remote Team Tech Overhaul

## 1.0 Executive Summary

The Field Operations and R&D teams faced systemic operational inefficiencies that created significant downtime, project delays, and safety risks. Core problems included a crippling lack of remote site visibility and unstable mobile connectivity for the field team, leading to an average of 4-6 hours of Mean Time To Resolve (MTTR) for network issues. Concurrently, the R&D lab suffered from constant stockouts of critical parts, causing approximately 20 hours of project downtime per month.

To address these challenges, a multi-pronged strategy was implemented using the Lean Six Sigma (DMAIC) methodology. The solution involved deploying a Unifi network and camera system for remote monitoring and safety, installing a Cradlepoint industrial cellular router to ensure reliable field connectivity, and rolling out the eMaint CMMS for automated inventory management.

This comprehensive overhaul successfully transformed the operational capabilities of both teams. The initiative resulted in a **95% reduction in downtime** from parts shortages, a **75% decrease in remote site troubleshooting time**, and a **100% reduction in reported safety incidents**.

## 2.0 KPI Improvement

| Metric | Before Project | After Project | Improvement |
|---|---|---|---|
| Downtime from R&D Parts Stockouts | ~20 hours/month | < 1 hour/month | **95% Reduction** |
| Mean Time To Resolve (MTTR) - Field Network Issues | 4-6 hours (incl. travel) | < 1 hour (remote diagnosis) | **75% Reduction** |
| Field Data Upload Success Rate (First Attempt) | ~55% (unstable hotspots) | >80% (stable connection) | **Vastly Increased Reliability** |
| Reported Safety Incidents / Near-Misses | 3 per year | 1 per year | **100% Reduction** |

## 3.0 Project Methodology: Lean Six Sigma (DMAIC)

**Phase 1: Define**
The organization faced systemic downtime and safety risks across its R&D and Field Operations teams. The core problems were defined as: a crippling lack of remote site visibility, unstable mobile connectivity, and constant R&D stockouts of critical consumables.

**Phase 2 & 3: Measure & Analyze**
Initial data confirmed the severity of the KPIs. To understand the root causes, a Fishbone (Ishikawa) Diagram was used to structure a brainstorming session with key stakeholders.

**Technology**
- Lack of remote visibility
- No remote power cycling
- Unreliable personal hotspots
- No centralized asset tracking

**Methods (Process)**
- Manual inventory counts
- Reliance on travel for diagnostics
- No process for remote support
- Inconsistent data access

**Materials**
- Fragile, non-industrial cabling
- Frequent stockouts of critical parts
- No visibility into stock levels
- Disorganized parts storage

**Operational Downtime & Inefficiency**

**Environment**
- Harsh industrial conditions
- Surface wires pose trip hazards
- Mobile trailer in varied locations
- Poor cell service in field

**People**
- Lack of remote supervision
- Time wasted searching for parts
- Frustration from tech failures
- Inconsistent safety adherence

**Measurement**
- No data on consumable usage
- Downtime not accurately tracked
- No metrics on network reliability
- Inability to forecast part needs

**Phase 4: Improve**
Based on the analysis, a multi-faceted project was executed: deployed a UniFi network and camera system for remote monitoring, implemented a Cradlepoint industrial router for reliable connectivity, and rolled out eMaint CMMS for automated inventory management.
**Phase 5: Control**
To sustain the gains, comprehensive Standard Operating Procedures (SOPs) were developed for all new systems, and the eMaint dashboard was established as the control tool for monitoring inventory levels and usage trends.

## Statement of Work (SOWs)

### SOW-001: R&D Center Remote Monitoring & Safety System

**Client:** Internal R&D / Field Operations

**1. Project Overview & Business Need**
This project will engineer and deploy a robust remote monitoring solution to increase personnel safety, reduce diagnostic-related downtime, and improve management oversight at the R&D test facility. The current lack of visibility creates unacceptable safety risks and operational inefficiencies.

**2. Scope of Work**
- **In Scope:** Installation of UniFi network controller, PoE switch, Wi-Fi access points, (5) UniFi G5 cameras; replacement of existing cabling with industrial-grade, shielded CAT6 and protective floor channeling; configuration of remote access portal; development of user and technician SOPs.
- **Out of Scope:** Physical server infrastructure changes; ongoing data subscription costs; integration with third-party software beyond standard UniFi capabilities.

**3. Key Deliverables**
1. Fully functional and tested UniFi Network and Protect system.
2. Live, secure camera feeds accessible via web portal.
3. Completed installation of all cabling and hardware.
4. Finalized SOP documents for viewing and troubleshooting.

### SOW-002: Mobile Operations Connectivity Solution (Cradlepoint)

**Client:** Field Operations Team

**1. Project Overview & Business Need**
The Field Operations team currently relies on unstable, consumer-grade personal hotspots, resulting in poor connectivity, failed data uploads, and significant lost time. This project will deploy an industrial-grade, dual-SIM cellular router to provide reliable, high-speed internet connectivity in the mobile trailer, enabling real-time data transfer and access to cloud services.

**2. Scope of Work**
- **In Scope:** Procurement and installation of a Cradlepoint IBR1700 router; installation of a high-gain, roof-mounted cellular antenna; activation and configuration of dual SIM cards (AT&T, Verizon) for failover; configuration of local Wi-Fi network; development of field troubleshooting SOP.
- **Out of Scope:** Negotiation of cellular data plans; support for personal devices connecting to the network; advanced network configuration (VPN, custom firewall rules).

**3. Key Deliverables**
1. Fully installed and configured Cradlepoint IBR1700 router.
2. Operational dual-SIM failover for maximum uptime.
3. A stable, secure Wi-Fi network within and around the mobile trailer.
4. Finalized SOP document for field team troubleshooting.

### SOW-003: Automated Inventory Management System (eMaint)

**Client:** R&D Operations Team

**1. Project Overview & Business Need**
The R&D lab experiences frequent and costly project delays due to stockouts of critical consumables. The lack of an inventory system prevents usage tracking, forecasting, and timely reordering. This project will implement the eMaint CMMS to automate inventory management, eliminating stockouts and providing valuable usage data.

**2. Scope of Work**
- **In Scope:** Configuration of the eMaint inventory module; data import of initial parts list; creation of a standardized bin labeling system; procurement and setup of a barcode scanner; development and delivery of user training; creation of SOPs for all key inventory processes.
- **Out of Scope:** Integration with accounting software; management of vendor relationships or purchasing; physical reorganization of the stock room beyond bin labeling.

**3. Key Deliverables**
1. A fully configured eMaint inventory database with all critical parts loaded.
2. A physically labeled and organized stock room corresponding to the eMaint system.
3. A functional barcode scanning system for check-in/check-out.
4. Training session completed with all relevant R&D personnel.
5. Finalized SOP documents for all inventory management functions.

## Standard Operating Procedures (SOPs)

### UniFi System SOPs

**SOP-VIEW-001: Remote Viewing & Export**
**Purpose:** To define the standard process for authorized personnel to access live camera feeds and export video evidence from the UniFi Protect system.

1. **Access Portal:** Navigate to **https://unifi.ui.com/** and log in. Select the appropriate site, then click the "Protect" icon.
2. **Live View:** The default "Live View" shows a grid of all cameras. Click any camera to view it full-screen. Use the digital zoom controls as needed.
3. **Find Recordings:** Click the "Detections" tab for a list of motion events, or the "Timelapse" tab to scrub through all footage chronologically. Use the calendar and camera filters to narrow your search.
4. **Export Clip:** When viewing a recording, click the "Download Clip" (scissors) icon. Drag the start and end handles on the timeline to select the precise portion needed. Click 'Download' to save the MP4 file to your computer.

**SOP-TECH-001: Field Troubleshooting**

**SAFETY FIRST:** Always ensure the main equipment is powered down and you have clearance before physically inspecting cables in active machinery areas.

1. **Initial Triage:** If a camera is offline, first perform a hard refresh of your browser (Ctrl+F5).
2. **System-Wide Outage:** Check the UniFi PoE Switch lights. The main status light should be SOLID BLUE. If the switch is on, perform a hard power cycle: unplug it, wait 60 seconds, plug it back in.
3. **Single Camera Outage:** Check the switch port light. SOLID BLUE is good, AMBER indicates a cable fault. Trace the cable for damage and use a tester to verify wiring. If the cable is good, attempt a physical reset of the camera.

### Cradlepoint System SOPs

**SOP-FIELD-004: Mobile Internet Troubleshooting**
**Purpose:** To empower field teams to resolve common internet connectivity issues with the Cradlepoint router, minimizing downtime.

1. **Step 1: The 60-Second Power Cycle (Fixes 90% of issues).** Unplug the router, wait a full 60 seconds, plug it back in. Wait 5 minutes for it to fully boot and connect before testing.
2. **Step 2: Check Status Lights.**
   - **Power:**
     SOLID GREEN is normal.
   - **Modem:**
     SOLID GREEN is connected.
     BLINKING GREEN is connecting.
   - **Signal Strength:**
     3-4 BLUE bars is excellent. 1-2 is weak.

- **Step 3: Check Antennas.** Ensure all antennas are screwed on finger-tight and pointing straight up.
- **Step 4: Escalation.** If issues persist, contact IT Support. Provide the status of the lights.

### eMaint Inventory SOPs

**SOP-INV-001: Issuing a Part to a Work Order**
**Purpose:** To correctly decrement inventory and assign costs by checking out parts against a specific job.

Scan Bin Barcode → Enter Quantity → Enter Work Order # → Click Save

**SOP-INV-002: Receiving a New Purchase Order**
**Purpose:** To correctly increment inventory when new stock arrives.

Enter PO Number → Verify Physical Count vs. Slip → Enter Quantity Received → Click Receive & Stock Item

**SOP-INV-003: Performing a Cycle Count**
**Purpose:** To regularly verify the accuracy of the system inventory against physical stock.

Generate Count Sheet → Physically Count Items in Bin → Enter Physical Count in System → Save & Report Discrepancies